4-17.

设两束电子分别为 a, b. 运动方程:

$$mn_0\left(\frac{\partial \vec{v}_{e1}}{\partial t} + \vec{v}\cdot\nabla\vec{v}_{e1}\right) = -en_0\vec{E}_1$$

引入波假设, E_1, v_{e1} n_{e1} $\propto \exp[i(kx-\omega t)]$. $\vec{k} = k_x\hat{x}$.

$$\frac{\partial}{\partial x} = ik, \quad \frac{\partial}{\partial t} = -i\omega$$

线性化,

$$m(-i\omega + ikv_0)\vec{v}_{e1} = -e\vec{E}_1$$

$$\vec{v}_{e1} = -\frac{ie}{m}\frac{E_1}{\omega - kv_0}$$

电子连续性方程:

$$\frac{\partial(n_0+n_{e1})}{\partial t} + \nabla\cdot[(n_0+n_{e1})(\vec{v}_0+\vec{v}_{e1})] = 0.$$

$$\Rightarrow \quad \frac{\partial n_{e1}}{\partial t} + \nabla\cdot(n_0\vec{v}_{e1} + n_{e1}\vec{v}_0) = 0$$

$$\Rightarrow \quad -i\omega n_{e1} + i\vec{k}(n_0\vec{v}_{e1} + n_{e1}\vec{v}_0) = 0.$$

$$n_{e1} = \frac{kn_0v_{e1}}{\omega - kv_0} = -\frac{iekn_0}{m}\frac{E_1}{(\omega-kv_0)^2}$$

$$n_{ej} = -n_{0j}\frac{ikE_1e}{m(\omega-kv_0)^2} \qquad (j = a, b).$$

泊松方程：

$$\nabla^2\varphi = -\frac{\rho}{\varepsilon_0} \Rightarrow \nabla\cdot\vec{E} = \frac{\rho}{\varepsilon_0} \qquad \rho = -e n_{e1},\ n_{e1} = n_{e1a} + n_{e1b}$$

Maxwell eq 线性化：

$$i\vec{k}\cdot\vec{E}_1 = \frac{\rho}{\varepsilon_0} = -\frac{e}{\varepsilon_0}(n_{e1a} + n_{e1b}).$$

对流 a，

$$v_{0a} = v_0\hat{x} \qquad n_{0a} = \frac{n_0}{2}$$

对流 b，

$$v_{0b} = -v_0\hat{x} \qquad n_{0b} = \frac{n_0}{2}$$

代入，有：

$$ikE_1 = \left(-\frac{e}{\varepsilon_0}\right)\left(-\frac{ikE_1 e}{m}\right)\left[\frac{n_0/2}{(\omega - kv_0)^2} + \frac{n_0/2}{(\omega + kv_0)^2}\right].$$

$$\Leftrightarrow \qquad 1 = \frac{e^2}{2\varepsilon_0}\frac{n_0}{m}\left[\frac{1}{(\omega - kv_0)^2} + \frac{1}{(\omega + kv_0)^2}\right].$$

又 $\omega_p^2 = \frac{e^2 n_0}{\varepsilon_0 m_e}$，

$$1 = \frac{1}{2}\omega_p^2\left[\frac{1}{(\omega - kv_0)^2} + \frac{1}{(\omega + kv_0)^2}\right].$$

色散关系.

(2).

$$\frac{1}{2}\omega_p^2\left[\frac{\omega^2+2\omega k v_0+k^2v_0^2}{[(\omega-kv_0)(\omega+kv_0)]^2}+\frac{\omega^2-2\omega kv_0+k^2v_0^2}{[(\omega-kv_0)(\omega+kv_0)]^2}\right]=1.$$

$$\Leftrightarrow \quad \omega_p^2\frac{\omega^2+kv_0^2}{(\omega^2-k^2v_0^2)^2}=1.$$

$$\Leftrightarrow \quad \omega^4-2k^2v_0^2\omega^2+k^4v_0^4-\omega_p^2\omega^2+k^2v_0^2\omega_p^2=0.$$

$$\Leftrightarrow \quad \omega^4-(2k^2v_0^2-\omega_p^2)\omega^2+k^2v_0^2(k^2v_0^2-\omega_p^2)=0.$$

求解：

$$\Delta=8k^2v_0^2\omega_p^2+\omega_p^4.$$

$$\omega^2=\frac{(2k^2v_0^2-\omega_p^2)\pm\sqrt{8k^2v_0^2\omega_p^2+\omega_p^4}}{2}$$

令 $x\equiv\frac{2k^2v_0^2}{\omega_p^2}$ $y^2\equiv\frac{2\omega^2}{\omega_p^2}$，代入：

$$y^2=1+x\pm(4x+1)^{\frac{1}{2}}.$$

$$\Leftrightarrow \quad y^2-x\mp(4x+1)^{\frac{1}{2}}=1.$$

求极小. 时间不稳定性

$$\frac{\partial F}{\partial x}=-1\mp\frac{1}{2}(4x+1)^{-\frac{1}{2}}\cdot 4=-1\mp\frac{2}{\sqrt{4x+1}}$$

令 $\frac{\partial F}{\partial x}=0$,

$$1=\mp\frac{2}{\sqrt{4x+1}},\quad x=\frac{3}{4}$$

又

$$\frac{\partial^2 F}{\partial x^2} = \pm 4(4x+1)^{-\frac{3}{2}}$$

when $x=\frac{3}{4}$，由于应为极小值，$\frac{\partial^2 F}{\partial x^2} < 0$. "±" 应取 "−".

此时，y 为复. 令 $y = i\gamma$. 代入：

$$\gamma^2 = (4x+1)^{\frac{1}{2}} - (x+1).$$

$$\gamma^2 = \left(\frac{8k^2v_0^2}{\omega_p^2}+1\right)^{\frac{1}{2}} + \left(1+\frac{2k^2v_0^2}{\omega_p^2}\right).$$

$$\frac{d\gamma^2}{dx} = 0 \quad \Rightarrow \quad 2(4x+1)^{1/2} - 1 = 0.$$

$$\Rightarrow \quad x=\frac{3}{4}. \quad \gamma_{max} = \frac{1}{2} = \frac{\sqrt{2}\,Im(\omega)}{\omega_p} \quad Im(\omega) = \frac{\omega_p}{2^{3/2}}$$

$$\frac{3}{4} = \frac{2k^2\omega^2}{\omega_p^2} \quad \Rightarrow \quad k = \frac{\sqrt{6}\omega_p}{4v_0^2}$$